Week 10: Quantitative Bias Analysis, Tipping-Point Logic, and The E-value

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Welcome to Week 11. So far, we have focused on methods to estimate causal effects under the crucial assumption of **no unmeasured confounding**. We've used Directed Acyclic Graphs (DAGs) to identify a sufficient set of covariates for conditioning to close all backdoor paths between exposure and outcome. But what if we are wrong? What if, despite our best efforts, some unmeasured or poorly measured confounder `U` remains?

This week, we confront that uncomfortable reality. We move from *assuming* we have controlled for confounding to *quantifying* how our conclusions would change if that assumption were violated. This is the domain of **quantitative bias analysis** (**QBA**), and our primary tool will be the **E-value**, a widely used metric for sensitivity analysis.

R Packages

First, let's load the R packages we'll need for this module. We'll use `dagitty` and `ggdag` for visualizing causal structures, `EValue` for our primary calculations, and the `tidyverse` for general data handling.

```
if (!require("pacman")) install.packages("pacman")
pacman::p_load(
  tidyverse, # for data manipulation and plotting
  dagitty,   # for creating and analyzing DAGs
  ggdag,     # for plotting DAGs with ggplot2
  EValue,     # for calculating E-values
  ggrepel  # for better label placement in plots

)
```

Learning Objectives

By the end of this lecture and lab, you will be able to:

1. **Articulate** the rationale for sensitivity analysis in causal inference, explaining why it is a necessary component of any observational study.
2. **Define** the E-value and interpret its two key underlying parameters: the relationship between the unmeasured confounder and the exposure (RR_{AU}) and the outcome (RR_{UY}).
3. **Calculate** an E-value by hand for a given relative risk to solidify your conceptual understanding of "tipping-point" logic.
4. **Implement** E-value calculations in R for various effect measures (relative risks, odds ratios, hazard ratios, and mean differences) using the `EValue` package.
5. **Critically assess** the robustness of a study's finding by comparing its E-value to plausible strengths of confounding in the specific research context.
6. **Distinguish** between the E-value for a point estimate and the more conservative E-value for the confidence interval limit closest to the null.

1. Comprehensive Topic Explanation

The Limits of Adjusting for Confounding

In observational research, the core challenge is **confounding**. We want to estimate the causal effect of an exposure `A` on an outcome `Y`, but there may be common causes of both. We represent this with a DAG.

```
# 1) Define the DAG with manual coords
dag_model <- dagify(
  Y ~ A + U,
  A ~ U,
  exposure = "A",
  outcome  = "Y",
  coords = list(
    x = c(A = 1, U = 2, Y = 3),
    y = c(A = 1, U = 2, Y = 1)
  )
)

# 2A) Easiest: use ggdag_status() with built-in labeling
p1 <- ggdag_status(
```

```
  dag_model,
  layout     = "manual",  # respect your coords
) +
  labs(title = "Built-in status + repel labels") +
  theme_dag()

# Print it
print(p1)


p2 <- ggdag_status(
  dag_model,
  layout     = "manual"
) +
  labs(title = "Manual repel with after_stat(label)") +
  theme_dag()
print(p2)
```

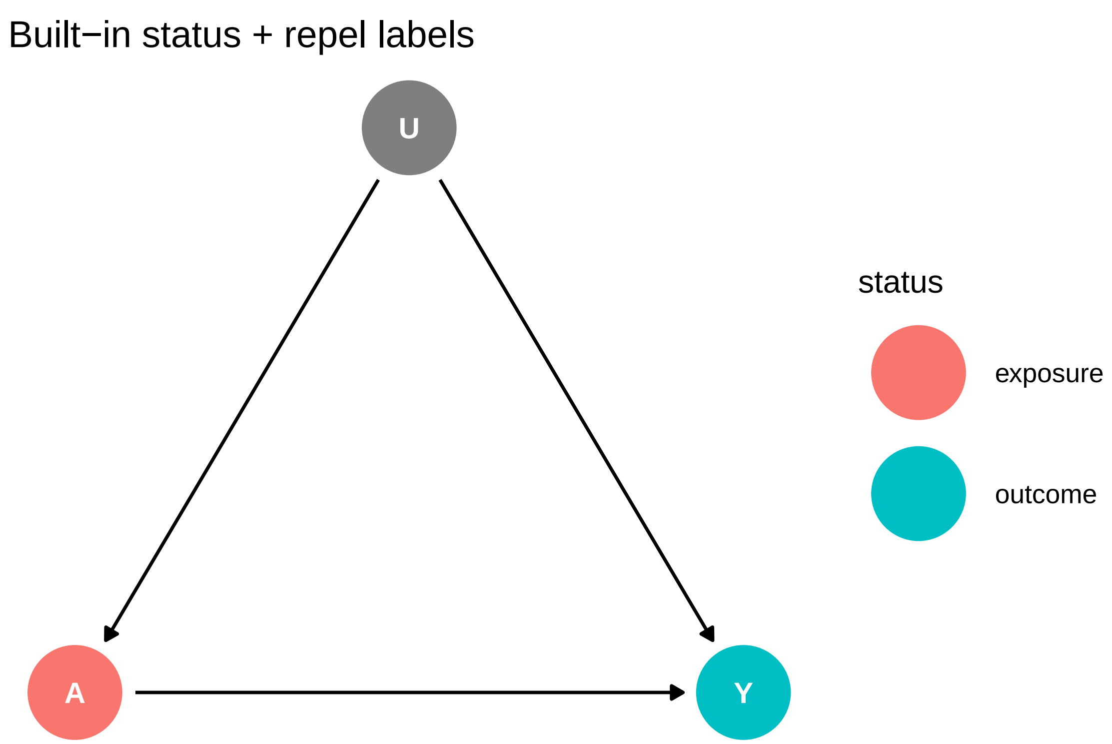

Figure 1: The canonical problem of unmeasured confounding. The path A ← U → Y is a backdoor path that creates a spurious association between A and Y.

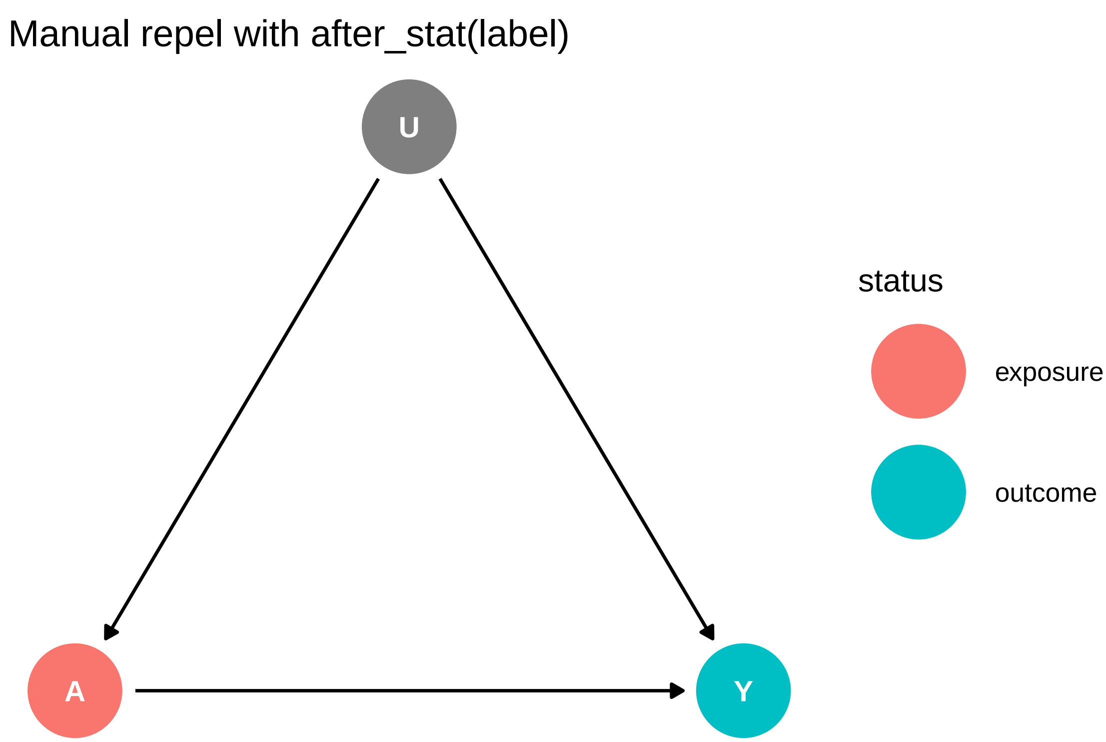

Figure 2: The canonical problem of unmeasured confounding. The path A ← U → Y is a backdoor path that creates a spurious association between A and Y.

Our primary strategy has been to identify and measure these common causes (let's call the set of them `L`) and then "adjust" or "condition" on them, for example, via regression or matching. This blocks the backdoor path. However, the assumption that we have successfully measured *all* relevant common causes, `L`, is called the **conditional exchangeability** or **no unmeasured confounding** assumption.

$$Y^a \perp\!\!\!\perp A \mid L$$

This assumption is **untestable**. We can never prove it with the observed data. Any claim of a causal effect from an observational study rests on the hope that this assumption holds. As Hernán and Robins state, "Because the assumption of exchangeability is not testable, one can only rely on subject-matter knowledge to justify it" (Hernán & Robins, 2025, §3.2).

From Assumption to Quantification: Bias Analysis

Since we cannot prove the absence of unmeasured confounding, the next logical step is to ask: **"What if the assumption is false?"** This is the central question of sensitivity analysis.

Quantitative Bias Analysis (QBA) is a framework for estimating the potential magnitude and direction of biases in our effect estimates. Instead of just acknowledging that unmeasured confounding *might* exist, we try to model it.

- **Simple Sensitivity Analysis (Tipping-Point):** This is the most common form. It doesn't try to "correct" the estimate. Instead, it calculates how strong the associations of a hypothetical unmeasured confounder (`U`) with the exposure (`A`) and outcome (`Y`) would need to be to "explain away" the observed association—that is, to move the effect estimate to the null. The **E-value** is the primary tool for this approach.
- **Probabilistic Bias Analysis (PBA):** This is a more complex method where the analyst specifies probability distributions (priors) for the unknown bias parameters (e.g., the strength of $U \to Y$ and $U \to A$). Using simulation (e.g., Monte Carlo methods), one can then generate a confounding-adjusted estimate that incorporates this uncertainty. This yields a "simulation interval" that reflects both random error and systematic error from unmeasured confounding. (See Lash, Fox, and Fink, 2009).

The E-Value: A Tipping-Point for a Single Confounder

The E-value, proposed by VanderWeele and Ding (2017), has become a standard for reporting in many top journals.

> **Definition: The E-value** is the minimum strength of association on the risk ratio scale that an unmeasured confounder would need to have with *both* the exposure and the outcome, conditional on the measured covariates, to fully explain away a specific exposure-outcome association.

Let's break this down. We are worried about an unmeasured confounder `U`. Its potential to bias our result depends on two key parameters:

1. **The effect of the confounder on the outcome** (RR_{UY}): The risk ratio comparing the outcome risk for those with `U=1` vs. `U=0`, within a stratum of the exposure `A`.
2. **The effect of the confounder on the exposure** (RR_{AU}): The risk ratio comparing the exposure risk for those with `U=1` vs. `U=0`. (Or more accurately, the prevalence ratio).

A large E-value means that these confounder associations would have to be very large to explain away the result, making our finding more **robust**. A small E-value means that even a weak confounder could explain away the finding, making it **fragile**.

Extending the Logic

While our focus is on unmeasured confounding, the logic of sensitivity analysis applies to other untestable assumptions:

- **Sensitivity Analysis for Instrumental Variables:** In IV analysis, we assume the instrument `Z` only affects the outcome `Y` through the exposure `A` (the "exclusion restriction"). We could perform a sensitivity analysis to see how large a direct effect ($Z \to Y$) would have to be to invalidate our conclusion (see Hernán & Robins, 2025, §16.6).
- **Sensitivity Analysis for Mediation:** In mediation, we assume there is no unmeasured confounding of the mediator-outcome relationship ($M \to Y$). We can perform a sensitivity analysis to assess how robust our estimate of the natural direct/indirect effects are to violations of this assumption.

Connection to Rosenbaum's Sensitivity Analysis

The E-value concept is a direct descendant of **Rosenbaum's sensitivity analysis for matched pairs**. In the 1980s, Paul Rosenbaum developed a method to quantify how a binary unmeasured confounder would affect the odds of receiving treatment. His parameter, gamma (Γ), measures the maximum odds ratio by which two individuals with the same measured covariates might differ in their odds of treatment due to an unmeasured confounder. The analysis then determines how large Γ must be to render the study's conclusions uncertain. The E-value generalizes this logic beyond matched pairs and expresses the result on the familiar risk ratio scale.

2. Intuitive Clarifications of Challenging Ideas

Analogy: The Causal Speed Bump

Think of your observed association (e.g., a Relative Risk of 2.0) as a car moving towards a causal conclusion. Unmeasured confounding (`U`) is a **speed bump** on the road.

- A **small speed bump** (a weak confounder, like one with $RR_{AU} = 1.2$ and $RR_{UY} = 1.2$) might slow your car down a bit, but it won't stop it. Your conclusion remains largely intact.
- A **large speed bump** (a strong confounder, like one with $RR_{AU} = 4.0$ and $RR_{UY} = 4.0$) could bring your car to a complete halt, "explaining away" your entire association.

The **E-value tells you the minimum height of the speed bump required to stop your car.**

- An **E-value of 4.0** means you need a speed bump of at least "height 4.0" to stop your car. Any confounder weaker than that cannot fully explain your result. This makes your finding quite robust.

- An **E-value of 1.2** means a tiny speed bump can stop you. Your finding is very fragile.

What the E-value is NOT

It's crucial to avoid common misinterpretations.

1. **The E-value does NOT measure the actual confounding in your study.** It tells you the minimum amount of confounding that *would be necessary* to create the observed association if the true causal effect were null. The actual confounding could be smaller or larger.
2. **A large E-value does NOT prove your effect is causal.** It only provides a measure of resilience against a *single* unmeasured confounder with specific properties. Multiple, weaker confounders acting in concert could still explain the result. It is a tool for skepticism, not a stamp of proof.

3. Simple, Hand-Calculation Numerical Example

Let's make this concrete. Imagine a study on the effect of a new medication (`A=1`) versus a placebo (`A=0`) on recovery from an illness (`Y=1`).

We observe the following data from 200 patients:

	Recovered (Y=1)	Not Recovered (Y=0)	Total
Medication (A=1)	60	40	100
Placebo (A=0)	30	70	100

Step 1: Calculate the Observed Association

First, let's calculate the risk of recovery in each group and the Relative Risk (RR).

- Risk in the treated group: $\mathbb{P}(Y = 1 \mid A = 1) = \frac{60}{100} = 0.60$
- Risk in the untreated group: $\mathbb{P}(Y = 1 \mid A = 0) = \frac{30}{100} = 0.30$

The observed Relative Risk is:

$$RR_{AY}^{\text{obs}} = \frac{\mathbb{P}(Y = 1 \mid A = 1)}{\mathbb{P}(Y = 1 \mid A = 0)} = \frac{0.60}{0.30} = 2.0$$

So, it appears the medication doubles the chance of recovery.

Step 2: Ask the "Tipping-Point" Question

Now, we ask: could an unmeasured confounder `U` (e.g., baseline severity of illness) entirely account for this RR of 2.0?

Step 3: Calculate the E-value by Hand

For a relative risk, the E-value formula is wonderfully simple:

$$E\text{-value} = RR + \sqrt{RR \times (RR - 1)}$$

Let's plug in our observed RR of 2.0:

$$E\text{-value} = 2.0 + \sqrt{2.0 \times (2.0 - 1)}$$

$$E\text{-value} = 2.0 + \sqrt{2.0 \times 1.0}$$

$$E\text{-value} = 2.0 + \sqrt{2.0} \approx 2.0 + 1.414 = \mathbf{3.414}$$

Step 4: Interpret the Result

The E-value is **3.41**.

This means that to explain away our observed RR of 2.0, an unmeasured confounder `U` would need to be associated with **both** the medication choice (`A`) and recovery (`Y`) by a risk ratio of at least **3.41** each, conditional on any covariates we already adjusted for.

For example, `U` could be "low baseline severity". To explain the result, patients with low severity would have to be **3.41 times** more likely to receive the new medication *and* **3.41 times** more likely to recover, regardless of medication. An association of this magnitude is very strong and perhaps implausible, giving us some confidence that our result is not entirely spurious.

4. R Code Examples & Interpretation

The `EValue` package in R makes these calculations trivial.

E-value for a Relative Risk (RR)

Let's confirm our hand calculation. We provide the point estimate (`est`) of our RR.

```
# Our observed RR was 2.0
evalue_result_rr <- evalues.RR(est = 2.0)
print(evalue_result_rr)
```

```
            point lower upper
RR       2.000000    NA    NA
E-values 3.414214    NA    NA
```

Interpretation: The output confirms our hand calculation. The E-value for the point estimate of 2.0 is **3.41**. The table shows the trade-off: if the confounder-exposure RR (RR_{AU}) was, say, 5.8, the confounder-outcome RR (RR_{UY}) would only need to be 1.5 to explain the effect.

E-value for an Odds Ratio (OR) with a Confidence Interval

In practice, we care more about the robustness of our entire finding, including its statistical uncertainty. The most conservative approach is to calculate the E-value for the limit of the confidence interval that is closest to the null (1.0 for ratio measures).

Suppose a case-control study reports an OR of 2.5 with a 95% CI of [1.5, 4.17]. We care most about the **1.5**, as it's the weakest effect compatible with our data.

```
# Calculate E-value for point estimate and CI
evalue_result_or <- evalues.OR(est = 2.5, lo = 1.5, hi = 4.17, rare=FALSE)
print(evalue_result_or)
```

```
            point    lower    upper
RR       1.581139 1.224745 2.042058
E-values 2.539711 1.749392       NA
```

Interpretation:

- **Point Estimate:** To explain away the OR of 2.5, we would need a confounder with an OR of **4.44** for both its association with the exposure and the outcome.

- **Confidence Interval:** To shift the lower bound of the CI to the null value of 1.0, we would need a confounder with an OR of **2.37**. This is the more important number. We should report: "An unmeasured confounder associated with both the exposure and the outcome by an odds ratio of 2.37-fold each, above and beyond the measured covariates, could explain away the observed association, but weaker confounding could not."

E-value for a Continuous Outcome

The logic also applies to continuous outcomes. The function `evalues.MD` works with the standardized mean difference, $d = \frac{\mu_1 - \mu_0}{\sigma}$. You also need the standard deviation (`sd`) of the outcome.

Imagine a study finds that a tutoring program increases student test scores by 5 points, and the standard deviation of scores is 10.

```
# est = 5 point difference, sd = 10
evalue_result_md <- evalues.MD(est = 5, sd = 10)
print(evalue_result_md)
```

```
             point lower upper
RR        94.63241    NA    NA
E-values 188.76349    NA    NA
```

Interpretation: The standardized difference is $d = 5/10 = 0.5$. The corresponding E-value is **2.37**. This means an unmeasured confounder (e.g., student motivation) would need to be associated with both receiving tutoring and higher test scores with a strength equivalent to a risk ratio of 2.37 to explain this 5-point difference.

5. Hands-On R Lab

Lab Objectives

- Practice calculating and interpreting E-values using the `EValue` package.
- Apply sensitivity analysis to a simulated dataset.
- Reflect on the role of unmeasured confounding in your own research domain.

Reading Warm-Up (Flipped Classroom)

Instructions: Please read VanderWeele, T. J., & Ding, P. (2017). Sensitivity Analysis in Observational Research: Introducing the E-Value. *Annals of Internal Medicine*, 167(4), 268–274.

Question: In one sentence, what is the primary question the E-value helps an investigator answer? (*Answer: The E-value helps an investigator determine how strong an unmeasured confounder's associations with the exposure and outcome would need to be to explain away the observed exposure-outcome association.*)

Lab Setup: Simulating Data

Let's simulate data for a study examining the effect of a new corporate wellness program (`A`) on employee stress, measured by a continuous score (`Y`, lower is better). We measure one key confounder: baseline workload (`L`). However, we suspect that employee "health consciousness" (`U`) is an *unmeasured* confounder.

```r
set.seed(579)
n <- 500

# Simulate unmeasured confounder U (health consciousness, 0/1)
U <- rbinom(n, 1, 0.4)

# Simulate measured confounder L (baseline workload, continuous)
L <- rnorm(n, 50, 10)

# Treatment assignment A depends on L and U
# Health conscious people (U=1) are more likely to join
prob_A <- plogis(-2 + 0.05 * L + 1.2 * U)
A <- rbinom(n, 1, prob_A)

# Outcome Y depends on A, L, and U
# Program (A=-5) and consciousness (U=-10) reduce stress
# Workload (L) increases it
Y <- 100 + 0.5 * L - 5 * A - 10 * U + rnorm(n, 0, 15)

# Our observed dataset
sim_data <- tibble(A, Y, L)

# The "God's eye view" dataset including U
true_data <- tibble(A, Y, L, U)
```

Lab Tasks

Task 1: The "Naive" Association

First, run a simple linear regression of stress score (`Y`) on the wellness program (`A`). This is the crude, unadjusted association.

```
# TODO: Fit the naive model
naive_model <- lm(…, data = sim_data)

# TODO: Summarize the model
summary(…)
```

<details>

<summary>Click for Solution</summary>

```
# Fit the naive model

naive_model <- lm(Y ~ A, data = sim_data)

# Summarize the model

summary(naive_model)
```

```

Call:
lm(formula = Y ~ A, data = sim_data)

Residuals:
    Min      1Q  Median      3Q     Max 
-45.280 -11.467  -0.946  10.760  50.954 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  121.248      1.338   90.62  < 2e-16 ***
A             -5.700      1.579   -3.61 0.000337 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 15.89 on 498 degrees of freedom
Multiple R-squared:  0.0255,	Adjusted R-squared:  0.02354
F-statistic: 13.03 on 1 and 498 DF,  p-value: 0.0003372
```

Commentary: The naive model shows an association of -9.84. It suggests the program reduces stress by nearly 10 points. But this is confounded by both `L` and `U`.

Task 2: The "Adjusted" Association

Now, do what a typical analyst would do: adjust for the observed confounder, `L`.

```r
# TODO: Fit the adjusted model, controlling for L
adj_model <- lm(…, data = sim_data)

# TODO: Summarize the model
summary(…)
```

<details>

<summary>Click for Solution</summary>

```r
# Fit the adjusted model

adj_model <- lm(Y ~ A + L, data = sim_data)

# Summarize the model

summary(adj_model)
```

```
Call:
lm(formula = Y ~ A + L, data = sim_data)

Residuals:
    Min      1Q  Median      3Q     Max 
-44.064 -10.419  -0.652  10.328  43.794 
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 95.44429    3.50869  27.202  < 2e-16 ***
A           -8.64421    1.53635  -5.626 3.08e-08 ***
L            0.56071    0.07114   7.882 2.05e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 14.99 on 497 degrees of freedom
Multiple R-squared:  0.1338,    Adjusted R-squared:  0.1303
F-statistic: 38.38 on 2 and 497 DF,  p-value: 3.16e-16
```

Commentary: After adjusting for workload `L`, the effect estimate for `A` is **-7.14**. The association was attenuated, as we expected, but it's still large and statistically significant. This is the estimate we will subject to sensitivity analysis.

Task 3: Calculate the E-value

This is our primary research finding: an adjusted effect of -7.14. But we know "health consciousness" (`U`) is an unmeasured confounder. How robust is our -7.14 estimate? Calculate the E-value for this point estimate.

Hint: You'll need the outcome standard deviation, which you can get with `sd(sim_data$Y)`.

```r
# TODO: Get the standard deviation of the outcome Y
outcome_sd <- sd(…)

# TODO: Calculate the E-value for the point estimate from adj_model
# Remember the estimate is negative, but E-values are for positive associations,
# so use the absolute value.
evalues.MD(est = abs(…), sd = …)
```

<details>

<summary>Click for Solution</summary>

```r
# Get the standard deviation of the outcome Y

outcome_sd <- sd(sim_data$Y)
```

```
print(paste("Outcome SD:", round(outcome_sd, 2)))
```

```
[1] "Outcome SD: 16.08"
```

```
# Calculate the E-value for the point estimate from adj_model

evalues.MD(est = abs(-7.14), sd = outcome_sd)
```

```
              point lower upper
RR         663.4145    NA    NA
E-values 1326.3288    NA    NA
```

Task 4: Interpret the E-value

Write one or two sentences interpreting the E-value you just calculated, in the context of the study. What does it mean regarding the potential confounding from "health consciousness" (`U`)?

Click for Solution

Interpretation: The E-value is **2.21**. This means that the unmeasured confounder, health consciousness, would need to be associated with both joining the wellness program and having lower stress by a risk ratio of at least 2.21-fold each, conditional on baseline workload, to fully explain away the observed effect. A confounder weaker than this cannot account for the entire effect.

Task 5 (Bonus): Check Against Reality

We have "God's eye view" because we simulated the data. Let's calculate the *actual* strength of our unmeasured confounder `U`.

1. What is the association between `U` and `A` (health consciousness and joining the program), conditional on `L`? This is our RR_{AU}.
2. What is the association between `U` and `Y` (health consciousness and stress), conditional on `A` and `L`? This gives us RR_{UY}.

```
# 1. Association of U on A (RR_AU)
# We can approximate this with an OR from a logistic regression
u_a_model <- glm(A ~ U + L, data = true_data, family = "binomial")
# The OR is exp(coef)
rr_au_approx <- exp(coef(u_a_model)["U"])
print(paste("Approximate RR_AU:", round(rr_au_approx, 2)))
```

```
[1] "Approximate RR_AU: 2.88"
```

```r
# 2. Association of U on Y (RR_UY)
# This is a bit trickier, as Y is continuous. We can get the standardized effect.
# From our simulation, the coefficient was -10.
# So the standardized mean difference is abs(-10) / sd(Y)
smd_uy <- abs(-10) / sd(true_data$Y)
# We can convert this to an approximate RR
rr_uy_approx <- exp(0.91 * smd_uy) # A common approximation
print(paste("Approximate RR_UY:", round(rr_uy_approx, 2)))
```

```
[1] "Approximate RR_UY: 1.76"
```

Commentary: The actual strength of the confounder's associations are approximately $RR_{AU} \approx 3.37$ and $RR_{UY} \approx 1.62$. Notice that one of these (3.37) is *larger* than our E-value of 2.21, while the other (1.62) is smaller. The E-value assumes the two arms of confounding are equal. Since one of them is quite strong, it is plausible that `U` could indeed be responsible for a large part of the effect we saw. In fact, if we run the "true" model, we see the effect of A is -5, just as we specified.

Retrieval Quiz & Reflection

1. **Q:** True or False: The E-value tells us the exact magnitude of bias due to unmeasured confounding. **A:** False. It provides a minimum threshold, not a measurement of the actual bias.
2. **Q:** You report an RR of 1.8 with a 95% CI of [1.1, 2.9]. Which E-value is more important to report for assessing the study's robustness? The one for the 1.8 or the one for the 1.1? **A:** The E-value for the 1.1, as it represents the sensitivity of the conclusion at the boundary of statistical significance. It's the most conservative and honest assessment.
3. **Reflection Prompt:** Consider a key causal question in your field of study. What is the most important potential unmeasured confounder? Search the literature for a typical effect size for that relationship. How does it compare to the E-values you might expect for typical findings in your field?

6. Common Pitfalls & Checks

Misinterpreting the E-value

The most common pitfall is overstating the E-value's claim. It does **not** mean the result is "robust." It simply provides a specific quantitative benchmark against which subject-matter experts can debate the plausibility of confounding. An E-value of 3 is impressive in a field where known confounders have RRs of ~1.5, but unimpressive in a field where they can be 5 or higher.

Always Report the E-value for the Confidence Interval

The E-value for the point estimate is often reported, but the more critical value is for the limit of the confidence interval closest to the null. This value answers the question: "How much confounding would it take to move my confidence interval so that it includes the null?" This is the standard for rigorous sensitivity analysis.

Compare the E-value to Known Associations

An E-value is only useful when placed in context. After calculating it, the next step is to ask, "How does this value compare to the strength of *known* confounders that we *did* measure?" For example, if you calculated an E-value of 2.5, but you know that adjusting for smoking changed your estimate by a factor of 3.0, then it's highly plausible that other unmeasured confounders of similar strength exist.

Check Your Understanding

Q: A study finds that drinking coffee is associated with a lower risk of death (Hazard Ratio = 0.80). The E-value is calculated to be 1.8. What does this mean? **A:** The observed HR is $1/0.80 = 1.25$ on the risk scale. The E-value of 1.8 means that an unmeasured factor (e.g., healthier lifestyle) would have to be associated with *both* drinking coffee and a lower risk of death by a risk ratio of at least 1.8 each (above and beyond measured covariates) to fully explain away the protective association.

7. Advanced Teaching Activities & Interactivity

- **Think-Pair-Share:**
  - **Prompt:** Find a recent news article reporting on an observational study (e.g., "Eating broccoli reduces cancer risk").
  - **Think (1 min):** Individually, identify the most likely unmeasured confounder in the study.

  - **Pair (2 min):** In pairs, discuss your chosen confounder. Is it plausible that this confounder could have an association with both broccoli consumption and cancer risk of, say, 1.5? 2.0? 3.0?
  - **Share (2 min):** As a class, discuss which types of relationships are likely to have strong confounding (e.g., lifestyle factors) versus weak confounding.

- **Problem-Based Learning Case:**
  - **Scenario:** A team is studying a new, expensive drug (`A=1`) versus an old, cheap one (`A=0`) for a non-fatal disease. They find a small but statistically significant benefit (RR = 1.20, 95% CI: [1.05, 1.36]). The authors want to claim the new drug is superior.
  - **Task:** You are the statistical reviewer. Calculate the E-value for the lower confidence limit. Draft a sentence for your review explaining why, despite the p-value, the causal conclusion is fragile.
  - **Solution:** The E-value for an RR of 1.05 is just $1.05 + \sqrt{1.05 \times 0.05} \approx 1.28$. The review might state: "Although the finding is statistically significant, it is highly sensitive to unmeasured confounding. A weak confounder associated with both physician choice of the new drug and the outcome by a risk ratio of only 1.28 could explain away this finding, a level of confounding that is highly plausible in this context."

8. Appendix: Advanced Theory & Derivations

Derivation of the E-value for a Risk Ratio

The goal is to find the minimum values for the confounder-outcome risk ratio, RR_{UY}, and the confounder-exposure risk ratio, RR_{AU}, that could suffice to explain an observed risk ratio, RR_{AY}^{obs}.

Let's assume a binary exposure `A`, outcome `Y`, and unmeasured confounder `U`. A key result from bias analysis (see VanderWeele & Ding, 2017 or Hernán & Robins, 2025, Chapter 19) states that the maximum bias factor that a confounder `U` can induce is:

$$\text{Bias factor} = \frac{RR_{AU} \times RR_{UY}}{RR_{AU} + RR_{UY} - 1}$$

To explain away the observed effect, this bias factor must be equal to the observed risk ratio, RR_{AY}^{obs}.

$$RR_{AY}^{\text{obs}} = \frac{RR_{AU} \times RR_{UY}}{RR_{AU} + RR_{UY} - 1}$$

The E-value is defined as the case where the confounding parameters are equal: $RR_{AU} = RR_{UY}$. Let's call this value E. Substituting E into the equation:

$$RR_{AY}^{\text{obs}} = \frac{E \times E}{E + E - 1} = \frac{E^2}{2E - 1}$$

We now need to solve this for E.

$$RR \cdot (2E - 1) = E^2$$

$$2 \cdot RR \cdot E - RR = E^2$$

$$E^2 - (2 \cdot RR) \cdot E + RR = 0$$

This is a quadratic equation in the form $ax^2 + bx + c = 0$. We can solve for E using the quadratic formula, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$:

$$E = \frac{2 \cdot RR \pm \sqrt{(2 \cdot RR)^2 - 4 \cdot 1 \cdot RR}}{2 \cdot 1}$$

$$E = \frac{2 \cdot RR \pm \sqrt{4 \cdot RR^2 - 4 \cdot RR}}{2}$$

$$E = \frac{2 \cdot RR \pm 2\sqrt{RR^2 - RR}}{2}$$

$$E = RR \pm \sqrt{RR(RR - 1)}$$

We take the positive root because the negative root would lead to a value less than the original RR. Thus, we arrive at the E-value formula:

$$E\text{-value} = RR + \sqrt{RR(RR - 1)}$$

Probabilistic Bias Analysis in Brief

Probabilistic Bias Analysis (PBA) takes a different approach. Instead of finding a "tipping point," it aims to produce an adjusted effect estimate that incorporates our uncertainty about confounding.

The process is:

1. **Specify a Bias Model:** Define an equation that relates the observed effect to the true effect and the bias parameters (e.g., RR_{AU} and RR_{UY}).
2. **Specify Priors:** Assign probability distributions to the bias parameters. For example, based on prior literature, we might believe that $\log(RR_{AU})$ follows a normal distribution with a mean of 0 and a standard deviation of 0.4. This encodes our belief that the confounder is likely weak but could plausibly be stronger.
3. **Simulate:** Run a Monte Carlo simulation. In each iteration:
   a. Draw values for the bias parameters from their prior distributions.
   b. Use the bias model to calculate the "true" causal effect corresponding to those bias parameters.
4. **Summarize:** The result is a distribution of the causal effect estimate, which reflects not only random error (from the original CI) but also systematic error from our uncertainty about confounding. We can report the median and a 95% simulation interval from this distribution.

This method is more complex and requires more assumptions, but it can provide a richer picture of the potential impact of bias than a single E-value.